3\. Vector Databases

Vector databases are specialized storage systems designed to store, index, and retrieve high-dimensional numerical representations called embeddings. Embeddings represent semantic information from text, images, audio, or other data types. Unlike traditional relational databases, which typically retrieve records through exact values or structured conditions, vector databases support similarity searches. Common similarity metrics include cosine similarity, Euclidean distance, and dot product. These databases are particularly useful in recommendation systems, semantic search, image retrieval, and retrieval-augmented generation applications. Popular vector database technologies include Milvus, Qdrant, Weaviate, and PostgreSQL with the pgvector extension.

4\. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is an architecture that combines information retrieval with large language models. Instead of relying exclusively on information stored in a model's parameters, a RAG system retrieves relevant documents from an external knowledge source and provides them as context during generation. A typical RAG pipeline consists of document ingestion, text extraction, chunking, embedding generation, vector storage, retrieval, and response generation. When a user submits a question, the system converts the question into an embedding and searches for semantically similar document chunks. The retrieved information is then included in the prompt sent to the language model. This approach allows applications to answer questions using private, domain-specific, or frequently updated information.